CHAPTER 1: Digital Systems and Binary Numbers

MULTIPLE CHOICE

1. Convert the hexadecimal number **CB8** to binary.
   a) 1100 1101 1011 b) 1100 1011 1000
   c) 1010 1011 1010 d) 1101 1001 1100

2. Convert the hexadecimal number **CB7** to octal.
   a) 6267 b) 6657
   c) 5468 d) 5647

3. Convert the binary number **11001101011** to hexadecimal.
   a) 5B6 b) $67C$
   c) 45A d) 66B

4. Convert the binary number **01101** to decimal.
   a) 23 b) 17
   c) 13 d) 18

5. Convert the octal number 125 to decimal.
   a) 107 b) 85
   c) 95 d) 70

6. Express the decimal number -25 as an 8-bit number in sign magnitude form
   A. 11100011 B. 10011101
   C. 11100010 D. 10011001

7. Convert the number $(111010110001)_2$ to hexadecimal
   A) EB1 B) FC1 C) 1BF D) FB1

8. Using 2's complement, 01101 –00101 equals ……..
   a) 01000 b) 10001
   c) 11000 d) 01101

9. What is the BCD number corresponding to the decimal value 72?
   A) 01110010
   B) 10010000
   C) 00100111
   D) 10010100

10. The lower and upper range for a 4-bit two's-complement number is:
    A) -64 and +63 B) -16 and +15 C) -32 and +31 D) -8 and +7

Answers: 1.B, 2.A, 3.D, 4.C, 5.B, 6.D, 7.A, 8.A, 9.A, 10.D